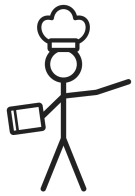
# The Kitchen CPU

Who does what in the fetch–decode–execute cycle

CPU



## Head Chef

**Control Unit (CU)**

Decodes each step with the codebook and gives the orders.



## Station Cook

**Arithmetic Logic Unit (ALU)**

Does the sums, like 12 + 30.

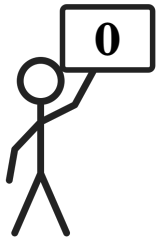


## Prep Shelf

**Cache**

Keeps copies of the lines used most, close to hand.

### Registers




## Line Counter

**Program Counter (PC)**

Knows which line comes next. Adds 1 after every fetch.

address



## Caller

**Memory Address Register (MAR)**

Holds the line number and calls it out to the Librarian.

address



## Bowl

**Accumulator (ACC)**

Sits on the scales. The reading is the running total.



## Scribe

**Memory Data Register (MDR)**

Copies down whatever the Librarian reads out: an instruction or a number.

data

## Librarian

**Memory (RAM)**

| Line | |
|---|---|
| **0** | `LOAD 5` |
| **1** | `ADD 6` |
| **2** | `STORE 7` |
| **3** | `STOP` |
| **4** | |
| **5** | `12 g` |
| **6** | `30 g` |
| **7** | `Total: ____` |

Keeps the recipe card, outside the CPU. A line's number is its address. Reading a line leaves it there.